Problem 1. We show that $X = \mathbb{R}^n \setminus \{0\}$ is path connected if and only if $n > 1$ (here $n \geq 1$).

Case $n = 1$. Then $X = (-\infty, 0) \cup (0, \infty)$, a union of two disjoint nonempty open sets, so X is not connected, hence not path connected. Directly: if $\gamma : [0, 1] \to X$ were a path from -1 to 1, the intermediate value theorem would give t with $\gamma(t) = 0 \notin X$, a contradiction.

Case $n \geq 2$. Let $x, y \in X$. Recall the segment $[x, y] = \{(1 - t)x + ty : t \in [0, 1]\}$ is the image of a path (a straight line path).

If $0 \notin [x, y]$, the straight line path from x to y lies in X.

Otherwise $0 = (1 - t)x + ty$ for some $t \in [0, 1]$. Since $x, y \neq 0$ we have $t \in (0, 1)$, so x and y are linearly dependent: $y = -cx$ with $c = \frac{1-t}{t} > 0$. Since $n \geq 2$, choose $z \in \mathbb{R}^n$ linearly independent from x (then also from y, as y is a nonzero multiple of x). In particular $z \neq 0$.

Claim: $0 \notin [x, z]$. If $0 = (1 - s)x + sz$ with $s \in [0, 1]$, then $s \neq 0$ (as $x \neq 0$) and $s \neq 1$ (as $z \neq 0$), so $z = -\left(\frac{1-s}{s}\right)x$, contradicting linear independence. The same argument shows $0 \notin [z, y]$.

Hence the segment from x to z followed by the segment from z to y is a path in X from x to y. So X is path connected. □

Problem 2.

$\mathbb{R}^2 \setminus \mathbb{Q}^2$ *is connected.* We show it is path connected, which implies connected. Let $x, y \in \mathbb{R}^2 \setminus \mathbb{Q}^2$ with $x \neq y$.

For a point $p \in \mathbb{R}^2$, consider the lines through p; they are parametrized by their direction angle $\theta \in [0, \pi)$, so there are uncountably many. For each $q \in \mathbb{Q}^2$ there is exactly one line through x and q (note $q \neq x$ since $x \notin \mathbb{Q}^2$). Since $\mathbb{Q}^2$ is countable, only countably many lines through x meet $\mathbb{Q}^2$. So all but countably many lines through x are contained entirely in $\mathbb{R}^2 \setminus \mathbb{Q}^2$. The same holds for y.

Choose a line L_x through x missing $\mathbb{Q}^2$. Then choose a line L_y through y missing $\mathbb{Q}^2$ with a direction different from that of L_x (possible, since uncountably many directions are available and only countably many are bad, and we exclude just one). Then L_x and L_y are not parallel, so they meet at a single point p. Go from x to p along L_x, then from p to y along L_y. This is a path in $\mathbb{R}^2 \setminus \mathbb{Q}^2$ from x to y. Hence $\mathbb{R}^2 \setminus \mathbb{Q}^2$ is path connected, and therefore connected.

$\mathbb{R} \setminus \mathbb{Q}$ *is not connected.* Let $U = (-\infty, 0) \cap (\mathbb{R} \setminus \mathbb{Q})$ and $V = (0, \infty) \cap (\mathbb{R} \setminus \mathbb{Q})$. Both are open in the subspace $\mathbb{R} \setminus \mathbb{Q}$ and they are disjoint. Since $0 \in \mathbb{Q}$, we have $0 \notin \mathbb{R} \setminus \mathbb{Q}$, so $U \cup V = \mathbb{R} \setminus \mathbb{Q}$. Both are nonempty ($-\sqrt{2} \in U$, $\sqrt{2} \in V$). So $\mathbb{R} \setminus \mathbb{Q}$ is the union of two disjoint nonempty open sets, hence disconnected. □

Problem 3. No, it is not necessary.

Counterexample. Let $X = [0, 1]$ and $Y = (0, 1)$, both with the subspace topology from $\mathbb{R}$.

- $f : X \to Y$, $f(x) = \frac{1}{4} + \frac{x}{2}$. This is continuous and injective, with image $\left[\frac{1}{4}, \frac{3}{4}\right]$, and the inverse $\left[\frac{1}{4}, \frac{3}{4}\right] \to [0, 1]$, $y \mapsto 2y - \frac{1}{2}$, is continuous. So f is a homeomorphism onto its image, i.e. an embedding.
- $g : Y \to X$, the inclusion $g(y) = y$. It is an embedding, since the topology on Y is the subspace topology inherited from $\mathbb{R}$, which is also the subspace topology inherited from X.

But X and Y are not homeomorphic: $X = [0, 1]$ is compact (Heine–Borel), whereas $Y = (0, 1)$ is not, since the open cover $\left\{\left(\frac{1}{n}, 1\right) : n \geq 2\right\}$ has no finite subcover (a finite subfamily has union

$\left(\frac{1}{N}, 1\right)$ for the largest $n = N$ used, which misses $\frac{1}{2N}$, a point of Y). Compactness is preserved by homeomorphisms, so $X \nsim Y$. $\square$

Problem 4. Let $\mathcal{B}$ be a base for the topology on X.

($\Longrightarrow$) If X is compact, then every open cover of X has a finite subcover. A cover consisting of elements of $\mathcal{B}$ is in particular an open cover (basis elements are open), so it has a finite subcover.

($\Longleftarrow$) Suppose every open cover of X consisting of elements of $\mathcal{B}$ has a finite subcover. Let $\mathcal{U}$ be an arbitrary open cover of X. For each $x \in X$ choose $U_x \in \mathcal{U}$ with $x \in U_x$, and since U_x is open and $\mathcal{B}$ is a base, choose $B_x \in \mathcal{B}$ with $x \in B_x \subseteq U_x$.

Then $\{B_x : x \in X\}$ is a cover of X (each $x \in B_x$) by elements of $\mathcal{B}$. By hypothesis, there are $x_1, ..., x_n \in X$ with $X = B_{x_1} \cup ... \cup B_{x_n}$. Since $B_{x_i} \subseteq U_{x_i}$, we get

$$X = U_{x_1} \cup ... \cup U_{x_n},$$

so $\left\{U_{x_1}, ..., U_{x_n}\right\}$ is a finite subcover of $\mathcal{U}$. Hence X is compact. $\square$